

Facilitator Guide

TAKE NOTE

This Manual should be used in conjunction with the Learner Guide and the Learner Workbook

Unit Standard Title: Demonstrate an understanding of the use of different number bases and measurement units and an awareness of error in the context of relevant calculations.

Unit Standard No: 9010

Unit Standard Credits: 2

NQF Level: 3

Time allocation

Training: The following breakdown can be used for the program.

Session: 1 = 30 min	Study unit	Day	Type
		Day___	
		Day___	
		Day___	
		Day___	
		Day___	
		Day___	
		Day___	
		Day___	
		Day___	
		Day___	
		Day___	
		Day___	
	Summative assessment	Day___	

Time allocation

Block program: The block program is only a guideline to indicate what will be facilitated . It is important to note that the block program can be changed depending on the pace of the learners and how many learners attended the learning program.

DAY	08:00 TO 08:30	08:40 TO 09:10	09:20 TO 10:00		10:30 TO 11:00	11:10 TO 11:40	11:50 TO 12:30		13:00 TO 13:30	13:40 TO 14:10	14:20 TO 15:00		15:15 TO 16:00
1				T E A				L U N C H				T E A	
2													
3													
4													
5													
6													
7													
8													
9													
10													
11													
12													
13													
14													

Learning event plan

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Use the format in table beneath to structure the content for a learning event. This provides guidelines for drawing up a facilitation plan in preparation for a presentation. An example of a facilitation plan is indicated in the next table.

Introduction	
Topic	♦ Select a suitable topic. ♦ Introduce the topic verbally. ♦ Display the topic visually.
Interest stimulus	♦ The aim of the interest stimulus is to obtain the learners' interest in the topic under discussion. ♦
Outcomes	♦ The outcome(s) state what the learners should be able to do after the learning event. ♦ Display the outcomes visually. ♦
Scope	♦ The scope of the presentation consists of all the main headings which will be covered in the theory lesson. ♦ Display the scope visually. ♦ Present the scope in a logical sequence. ♦
Importance	♦ The importance of the theory lesson must be stressed so that the learners realize how they will benefit by attending it. ♦
Ground rules	♦ The aim of establishing ground rules is to allow the learners to produce their own set of rules. ♦ Allow the learners to commit themselves to the ground rules by doing an observable activity. ♦
Physical environment	♦ The physical environment must be effective for learning. It includes aspects such as seating arrangement and equipment. ♦

Contents (Body)

Scope point no 1

The first scope point of the presentation will now be discussed in detail. Cover the rest of the scope points one by one.



- Establish a positive relationship with the learners (e.g. professional behavior).
- Indicate throughout the presentation which scope point is being discussed.
- Use at **least two relevant learning aids** (e.g. video, posters).
- The learning aids must support the theory.
- Present the theory lesson on the level of the target group.
- Explain the theory component by making use of appropriate examples.
- Allow the learners to ask questions or to comment.
- Apply effective verbal communication skills (e.g. speak loud enough, vary tone of voice).
- Apply effective non verbal communication skills (e.g. avoid irritating gestures, keep eye contact).
- Apply effective micro-skills (e.g. active listening, paraphrasing, summarizing).
- Control breathing so that voice usage is not affected by shortness of breath.
- Obtain maximum learner participation by including activities and group discussions.
- Keep control over disruptive influences (e.g. learners are not allowed to make unsuitable remarks and comments).
- Allow time for learners to ask questions during the presentation.

Closure

Summary

Questions

Assess

Interest stimulus

Summary

- ♦ Ask questions that are relevant to the outcomes stated.
 - ♦ Use different types of questions (e.g. close-ended, open-ended).
 - ♦ Avoid the use of questions such as leading, subjective, double-barreled etc.
 - ♦ Direct the questions to all the learners, instead of individuals.
 - ♦ Reinforce the questions and answers of the learners.
 - ♦ Provide positive feedback in time.
 - ♦ Allow activities to be performed to enable learners to reflect on the content.
 - ♦ Apply the theory to the practical situation (e.g. giving an example).
-
- ♦ Assess the learners by asking well selected questions concerning all the important aspects.



(It is very important that these questions are formulated in your planning phase and written into your presentation plan. It is thus determined whether the learners have achieved the outcome(s) of the theory lesson.)

- ♦ If there is a forthcoming presentation an interest stimulus must also be given to stimulate the interest of the learners for the next theory lesson.

Learning event plan

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Study unit: _____				
Learning Event: _____				
Topic: _____				
Series No	Facilitator activities	Learner activities	Learning aids	Time
	INTRODUCTION			

Learning event plan

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Series No	Facilitator activities	Learner activities	Learning aids	Time
	Interest stimulus:			
	Outcomes: ♦ ♦ ♦ ♦ ♦			
	Scope:			
	Importance			

Learning event plan

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Series No	Facilitator activities	Learner activities	Learning aids	Time
	Ground Rules			
	Scope point no 1: (Contents)			

Series No	Facilitator activities	Learner activities	Learning aids	Time
	Scope point no 2: (Contents)			

Series No	Facilitator activities	Learner activities	Learning aids	Time
	Scope point no 3: (Contents)			

Series No	Facilitator activities	Learner activities	Learning aids	Time
	Scope point no 4: (Contents)			

Series No	Facilitator activities	Learner activities	Learning aids	Time
	Scope point no 5: (Contents)			

Learning event plan

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Series No	Facilitator activities	Learner activities	Learning aids	Time
	Summary:			

Learning event plan

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Series No	Facilitator activities	Learner activities	Learning aids	Time
	Assess:			
	Interest stimulus:			

Program evaluation questionnaire

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Name of Facilitator: _____

Date: _____

Please give your honest reaction and comments. Mark the appropriate answer with an X

		Disagree	Neutral	Agree
LEARNING PROGRAM				
1.	The learning material is neatly compiled.			
2.	I received all the learning material in time.			
3.	The module content addressed my learning needs.			
4.	The exercises in the module enabled me to understand the theory.			
5.	The self assessments in the module enabled me to reach the stated outcomes.			
6.	The written assignment enabled me to master the content.			
7.	I could apply the theory into my work environment.			
LEARNING ENVIRONMENT				
8.	The venues were adequately prepared.			
9.	Suitable learning aids were used.			
LEARNING FACILITATORS				
10.	The learning facilitators demonstrated courteous behavior during training.			
11.	The learning facilitators were supportive towards me.			
12.	The learning facilitators appeared knowledgeable on the subject.			
13.	The learning facilitators demonstrated effective facilitation skills.			
14.	Enough time was allocated for me to practice facilitation skills.			
15.	I received adequate feedback on the progress that I made.			

Program evaluation questionnaire

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PROGRAM CONTENT										
		1	2	3	4	5	6	7	8	9
16.	How do you rate the subject content?									
17.	How do you find the quality of this program to be?									
18.	Rate the program for the qualities listed:									
	• Practical value									
	• Thoroughness									
	• New ideas gained									
	• Maintain your interest									
	• Helpful to developing yourself									
	• Clear, understandable.									
	• Did you understand the outcomes of the program?									

GENERAL COMMENTS BY LEARNER:

ACTION TAKEN BY FACILITATOR:

LEARNER GUIDE

Unit Standard Title:	Demonstrate an understanding of the use of different number bases and measurement units and an awareness of error in the context of relevant calculations.
Unit Standard No:	9010
Unit Standard Credits:	2
NQF Level:	3

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UNIT STANDARD 9010: Demonstrate an understanding of the use of different number bases and measurement units and an awareness of error in the context of relevant calculations.



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Publications disclaimer

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This material has been compiled based on the **UNIT STANDARD 9010: Demonstrate an understanding of the use of different number bases and measurement units and an awareness of error in the context of relevant calculations.**

The program has been accredited by SASSETA in the Republic of South Africa.

Material compiled by:

Training Dynamics Africa

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UNIT STANDARD 9010: Demonstrate an understanding of the use of different number bases and measurement units and an awareness of error in the context of relevant calculations.



1. Introduction

This learning program is part of a complete qualification. For more details concerning the complete qualification please contact your service provider or alternatively the office of the SASSETA

2. Purpose of this learning program

This unit standard is designed to provide credits towards the mathematical literacy requirements of the NQF at level 2. The essential purposes of the mathematical literacy requirements are that, as the learner progresses with confidence through the levels, the learner will grow in:

- ♦ An insightful use of mathematics in the management of the needs of everyday living to become a self-managing person.
- ♦ An understanding of mathematical applications that provides insight into the learner's present and future occupational experiences and so develop into a contributing worker.
- ♦ The ability to voice a critical sensitivity to the role of mathematics in a democratic society and so become a participating citizen.

People credited with this unit standard are able to:

- ♦ Convert numbers between the decimal number system and binary number system.
- ♦ Work with numbers in different ways to express size/magnitude.
- ♦ Demonstrate the effect of error in calculations.

2.1 Target group

This program is compiled for the following **target group**:

- ♦ Security members
- ♦ South African Defense Force members
- ♦ South African Police Force members
- ♦ Correctional Services
- ♦ Individuals who wishes to complete the NQF level 3 National Certificate in security practices.

3. Standards and qualifications

It is important that you as the learner understand and realize that this training program is not presented in isolation, but that it was developed based upon nationally recognized standards known as unit standards.

Unit standards are the "building blocks" of qualifications. All qualifications are plotted on the National Qualifications Framework (NQF).

Unit standards comprises of outcomes. An outcome is a statement that describes the required competency that must be demonstrated by the learner on successful completion of a training intervention.

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4. Assessments

In order to assess whether a learner can actually demonstrate the desired outcomes, assessment criteria are included in the unit standard. Each outcome has its own set of assessment criteria.

REMEMBER.....

The assessment criteria describe the evidence that is needed that will show that the learner has demonstrated the outcome correctly.

Kindly refer to the unit standard attached hereto for the assessment criteria.

It is of utmost importance that the learner fully understands the assessment criteria as listed in the unit standard, as it is the only way in which the learner will know what he will be assessed against.

The **final or summative assessment** is the most important aspect of this training program. It is during this process that the learner will be declared competent or not yet competent.

The learner will know exactly how he will be assessed, and when and where he will be assessed. All of these details must be obtained from the training provider where the learner enrolled for his program.

5. Range statements

REMEMBER.....

Also included in the unit standard are the range statements in support of the assessment criteria.

The range statements indicate detailed requirements of the assessment criteria.

6. The learning guide

The learner guide is included in this material under various learning units. The learner guide has been designed in such a manner that the learner is guided in a logical way through the learning material and requirements of the unit standard.

On completion of this skill program, the learner will be assessed against the assessment criteria as stipulated in the unit standard.

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Each of the modules consists of learner outcomes to be achieved during training and development interventions, work related experience or at your own time. The learner outcomes will enable you to achieve the required level of confidence and competence in order to undergo the summative (final evaluations) assessment.

The learner needs to assess his own knowledge and skill throughout the training process by completing the learner workbook. With the completion of the workbook you start collecting evidence to proof your competence. The learner workbook will be assessed (FORMATIVE ASSESSMENT)

TAKE NOTE

The learner guide will remain the property of the learner once the LEARNING PROGRAM has been completed.

TAKE NOTE

- The best results will be obtained if you start with Study Unit 1 in the Learner Guide, and work your way through it, study unit by study unit.
- All learning outcomes are vital and must be studied (and exercised for all practical requirements) thoroughly to ensure that enough evidence for all specific outcomes is generated.

7. Learning assumed to be in place

It is assumed that a learner which registers for this program have completed a program in communication at NQF Level 3 or equivalent.

It is imperative to understand that this program are only one of the building blocks to complete Skills program 1 The complete list of unit standards for Skills program appears on the next page.

7.1 Security program matrix

UNIT STANDARDS:

	ID	UNIT STANDARD TITLE	LEVEL	CREDITS
Core	244176	Use security equipment	Level 2	2
Core	114941	Apply knowledge of HIV/AIDS to a specific business sector and a workplace	Level 3	4
Core	244184	Apply legal aspects in a security environment	Level 3	8
Core	244177	Conduct a security patrol in an area of responsibility	Level 3	7
Core	117705	Demonstrate knowledge of the Firearms Control Act 2000 (Act No 60 of 2000) applicable to possessing a firearm	Level 3	3
Core	246694	Explain the requirements for becoming a security service provider	Level 3	4
Core	244182	Give evidence in court	Level 3	4
Core	244179	Handle complaints and problems	Level 3	6
Core	244181	Perform hand over and take over responsibilities	Level 3	2
Core	244189	Conduct access and egress control	Level 4	7
Core	242825	Conduct evacuations and emergency drills	Level 4	4
Core	11505	Identify, handle and defuse security related conflict	Level 4	12
Fundamental	119472	Accommodate audience and context needs in oral/signed communication	Level 3	5
Fundamental	9010	Demonstrate an understanding of the use of different number bases and measurement units and an awareness of error in the context of relevant calculations	Level 3	2
Fundamental	9013	Describe, apply, analyse and calculate shape and motion in 2- and 3-dimensional space in different contexts	Level 3	4
Fundamental	119457	Interpret and use information from texts	Level 3	5
Fundamental	9012	Investigate life and work related problems using data and probabilities	Level 3	5
Fundamental	119467	Use language and communication in occupational learning programmes	Level 3	5
Fundamental	7456	Use mathematics to investigate and monitor the financial aspects of personal, business and national issues	Level 3	5
Fundamental	119465	Write/present/sign texts for a range of communicative contexts	Level 3	5
Elective	117889	Demonstrate an understanding of Victim Empowerment	Level 2	6
Elective	12484	Perform basic fire fighting	Level 2	4
Elective	13912	Apply knowledge of self and team in order to develop a plan to enhance team performance	Level 3	5
Elective	244178	Apply neighbourhood watch principles	Level 3	5
Elective	113852	Apply occupational health, safety and environmental principles	Level 3	10
Elective	243188	Care for a service dog	Level 3	15
Elective	116534	Carry out basic first aid treatment in the workplace	Level 3	2

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Elective	113909	Coach a team member in order to enhance individual performance in work environment	Level 3	5
Elective	114942	Describe how to manage reactions arising from a traumatic event	Level 3	2
Elective	243190	Handle a trained service dog to deter crime	Level 3	8
Elective	119649	Handle and use a handgun	Level 3	2
Elective	119651	Handle and use a manually operated rifle or carbine	Level 3	2
Elective	119650	Handle and use a self-loading rifle or carbine	Level 3	2
Elective	119652	Handle and use a shotgun	Level 3	2
Elective	114979	Operate a computer workstation in a business environment	Level 3	2
Elective	244335	Conduct security at an event	Level 4	5
Elective	252174	Demonstrate an understanding of security measures within a port security environment	Level 4	4
Elective	246693	Explain the requirements for safeguarding a national key point	Level 4	4
Elective	123515	Handle and use a handgun for business purposes	Level 4	3
Elective	123519	Handle and use a manually operated rifle or carbine for business purposes	Level 4	3
Elective	123511	Handle and use a self loading rifle or carbine for business	Level 4	3
Elective	123514	Handle and use a shotgun for business purposes	Level 4	3
Elective	11985	Inspect, drive and maintain an official vehicle	Level 4	3
Elective	11513	Operate effectively within a specified control room environment	Level 4	15

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8. Preparation of summative assessment

After you have worked through the **LEARNER GUIDE** and you are satisfied that all theoretical and practical requirements/ evidence, as stipulated by the UNIT STANDARD, can be met, you need to make your own necessary arrangements with the assessor and/or facilitator. The training establishment will then arrange a suitable date for you to attend your pre-assessment meeting and then arrange for a suitable date to complete the final assessment.

8.1 RPL assessment



The assessment of RPL learners will be conducted in the same way as for those of new learners. The assessment pack is exactly the same and will therefore be used for new learners as well as RPL Learners. It must however be noted that learners who are applying for RPL must provide proof of previous learning and subject related experience prior to the assessment.

This proof or evidence can be in the format of certified copies (certificates) of previous learning programs that have been attended.

All the evidence will be assessed and authenticated before a learner will be allowed to enroll for an RPL program.

9. Roles and responsibilities

9.1 Facilitator/instructor

- Prepare to facilitate.
- Transfer of knowledge and skills

9.2 The role of the assessor

- Plan for the assessment
- Prepare the learner for the assessment

9.3 The role of the Learner

- Prepare for assessment.
- Accept the responsibility of learning.
- Accept the responsibility to produce/deliver evidence.

STUDY UNIT 1

Number systems and conversions

LEARNING OUTCOMES

On completion of this study unit the student will....



- ♦ Represent numbers in different ways.
- ♦ Round different types of numbers.
- ♦ Work with rational and irrational numbers and proper and improper fractions.
- ♦ Perform arithmetic with fractions.
- ♦ Calculate error ranges that result from arithmetic with rounded numbers.
- ♦ Estimate the correct answer to an arithmetic problem.
- ♦ Convert numbers between the decimal and binary systems.
- ♦ Perform addition and subtraction with binary numbers.
- ♦ Understand and work with the SI system of units.
- ♦ Understand the relationship between the SI system and other number systems.
- ♦ Estimate sizes of objects in the SI and UK systems.

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1. Number Bases

DEFINITION

In mathematical numeral systems, the base is usually the number of unique digits, including zero, that a positional numeral system uses to represent numbers.

EXAMPLE:

For example, for the decimal system (the most common system in use today) the base is 10, because it uses ten (10) digits: 0, 1, 2, 3, 4, 5, 6, 7, 8, 9.

In the binary system the base is 2 - it only uses two (2) digits: 1 and 0.

A digit's value is the digit multiplied by the value of its place. This is more commonly known as place value.

When we first learned about numbers, we were taught that, in the decimal system, things are organized into columns:

t	h	t	u	
2	1	9	3	"T" is the thousands column, "h" is the hundreds column, "t" is the tens column, and "u" is the units column.

So the number "2193" is 2-thousands plus 1-hundreds plus 9-tens plus 3-ones.

Years later, we learned that the ones column meant 10^0 , the tens column meant 10^1 , the hundreds column 10^2 , the thousands column 10^3 and so on, such that:

10^3	10^2	10^1	10^0	
2	1	9	3	The number 2193 is really: $\{(2 \times 10^3) + (1 \times 10^2) + (9 \times 10^1) + (3 \times 10^0)\}$

In any ordinary number that we are familiar with (i.e. a number from the decimal system or base 10), the value of each digit is determined by the position in the number that digit has, e.g.:

Let us write **four thousand five hundred and thirty one** - 4 531

EXAMPLE:

Thousands	Hundreds	Tens	Units
1 000's	100's	10's	1's
10^3	10^2	10^1	10^0
4	5	3	1
4×1000	5×100	3×10	1×1
= 4 531			

TAKE NOTE

Notice that each place value is 10 times larger than the place value on its right.

Seeing that we have a pretty good idea of how the decimal system works, let's take a more detailed look at the binary system and try and understand how it works.

2. The Binary System

There are several ways in nature that one can create binary numbers, e.g. a switch is open or closed, and a magnet has a north and a south pole. The main device currently being used is a transistor, which has two possible states: ON or OFF. The ON state is usually represented by a 1 and the OFF state by a 0.

Today's advanced technology is built on circuits and tiny electric pulses of 1's (ON) and 0's (OFF). All programs and data are recognised as just patterns of 1's and 0's by the digital computer. This system is called the Binary System (bi = two). 1 and 0 are known as binary digits, which in computer language have been shortened to bits.

In the decimal system, each digit's value is determined by it's position in the number, i.e. the place value.

The binary system works under the exact same principles as the decimal system, only it operates in base 2 rather than base 10. In other words, instead of columns being:

10^3 | 10^2 | 10^1 | 10^0 they are: 2^3 | 2^2 | 2^1 | 2^0

EXAMPLE:

If you're not used to them, binary numbers look pretty strange. Here's an example:

1 0 1 0 1 1 1 0⁴

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3. Converting from Binary to Decimal

So what is this number (1 0 1 0 1 1 1 0⁴) in the decimal system?

In the binary system, just two digits are used: 0 and 1. Therefore, any number must be represented by 0's and 1's only.



The first thing to know is that you read binary numbers from right to left.

The second thing you need to understand is that each digit is based on a power of the number 2:

- ♦ 2 to the power of 0 equals 1
- ♦ 2 to the power of 1 equals 2
- ♦ 2 to the power of 2 equals 4
- ♦ 2 to the power of 3 equals 8
- ♦ 2 to the power of 4 equals 16
- ♦ 2 to the power of 5 equals 32
- ♦ 2 to the power of 6 equals 64
- ♦ 2 to the power of 7 equals 128

See the pattern? Now let's take these numbers to use them as a template while remembering that binary numbers are read from right to left:

2 ⁷	2 ⁶	2 ⁵	2 ⁴	2 ³	2 ²	2 ¹	2 ⁰
128's	64's	32's	16's	8's	4's	2's	1's

Now let's use this template on that binary number from our earlier example. At the top is our template, at the bottom is our binary number:

28 4 2 6

Now we use simple multiplication and addition. If the binary number is a 1, it means this digit is "on" or "true" and we add the corresponding number from the template, if it is a 0, it means the digit is "off" or "false", and we do not add the corresponding number from the template.

In our example, the digits for 128, 32, 8, 4 and 2 are true, so we add $28+32+8+4+2 = 174$

You could also express it as: $28 \times 1 + 64 \times 0 + 32 \times 1 + 16 \times 0 + 8 \times 1 + 4 \times 1 + 2 \times 1 + 1 \times 0 = 174$

This means our binary number 10101110 is the number 174 in the decimal system.

The place values are powers of 2, with each place value being 2 times larger than the place value on it's right.

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2^7	2^6	2^5	2^4	2^3	2^2	2^1	2^0
128's	64's	32's	16's	8's	4's	2's	1's

Here are some decimal numbers represented in binary:

Decimal	Binary
1	1
2	10
3	11
4	100
5	101
6	110
7	111
8	1000
9	1001
10	1010

Try converting these numbers from binary to decimal:

- ♦ 10^4
- ♦ 111^4
- ♦ 1010^4
- ♦ 11110^4



2^4	2^3	2^2	2^1	2^0
16's	8's	4's	2's	1's
			1	0
		1	1	1
1	0	1	0	1
1	1	1	1	0

Solution:

- ♦ $10^4 = 2+0 = 2$
- ♦ $111^4 = 4+2+1 = 7$
- ♦ $1010^4 = 16+0+4+0+1 = 21$
- ♦ $11110^4 = 16+8+4+2+0 = 30$

4. Converting from Decimal to Binary

To convert a number from decimal to binary, you need to divide by 2 continuously until you cannot divide anymore. The remainder of each division is recorded.

EXAMPLE:

1

Convert 13 to binary.

$$\begin{array}{r}
 2 \overline{)13} \\
 2 \overline{)6} \text{ rem } 1 \\
 2 \overline{)3} \text{ rem } 0 \\
 1 \overline{)1} \text{ rem } 1 \\
 \hline
 = 1101^4
 \end{array}$$

EXAMPLE:

2

Convert 120 to binary.

$$\begin{array}{r}
 2 \overline{)120} \\
 2 \overline{)60} \text{ rem } 0 \\
 2 \overline{)30} \text{ rem } 0 \\
 2 \overline{)15} \text{ rem } 0 \\
 2 \overline{)7} \text{ rem } 1 \\
 2 \overline{)3} \text{ rem } 1 \\
 1 \overline{)1} \text{ rem } 1 \\
 \hline
 = 11111000^4
 \end{array}$$

Alternatively, you may prefer to use the following method:

EXAMPLE:

3

Convert 13 to binary.

What is the highest power of 2 which is less than or equal to 13? $2^3 = 8$

How much is left? $13 - 8 = 5$

What is the highest power of 2 less than or equal to 5? $2^2 = 4$

How much is left? $5 - 4 = 1$
 $= 2^0$

∴ $1 \times 2^3 + 1 \times 2^2 + 0 \times 2^1 + 1 \times 2^0$

$= 1101^4$

5. Binary addition

Binary numbers are added from right to left in the same way that decimal numbers are added.

5.1 Rules for binary addition:

- ♦ $0 + 0 = 0$
- ♦ $0 + 1 = 1$
- ♦ $1 + 0 = 1$
- ♦ $1 + 1 = 10$ (the 1 is carried to the next column).

EXAMPLE:

1

$$\begin{array}{r}
 1100^4 \\
 + 1011^4 \\
 \hline
 10111^4
 \end{array}$$

EXAMPLE:

2

$$\begin{array}{r}
 1001^4 \\
 + 1101^4 \\
 \hline
 10110^4
 \end{array}$$

6. Binary subtraction

Binary numbers are subtracted from right to left in the same way that decimal numbers are subtracted.

6.1 Rules Binary subtraction

- ♦ $0 - 0 = 0$
- ♦ $0 - 1 = 1$ borrow 1
- ♦ $1 - 0 = 1$
- ♦ $1 - 1 = 0$

EXAMPLE:

1

$$\begin{array}{r} 1110^4 \\ + 100^4 \\ \hline 1010^4 \end{array}$$

EXAMPLE:

2

$$\begin{array}{r} 1001^4 \\ 100^4 + \\ \hline 101^4 \end{array}$$

EXAMPLE:

3

$$\begin{array}{r} 10000^4 \\ + 101^4 \\ \hline 1011^4 \end{array}$$

7. Scientific Notation

Scientific Notation was developed in order to easily represent numbers that are either very large or very small.

Here are two examples of large and small numbers. They are expressed in decimal form instead of scientific notation to help illustrate the problem:

EXAMPLE:

- ♦ The Andromeda Galaxy (the closest one to our Milky Way galaxy) contains at least 200,000,000,000 stars.
- ♦ On the other hand, the weight of an alpha particle, which is emitted in the radioactive decay of Plutonium -239, is 0.000,000,000,000,000,000,000,006,645 kilograms.

As you can see, it could get tedious writing out those numbers repeatedly. So, a system was developed to help represent these numbers in a way that was easy to read and understand: Scientific Notation.

Using one of the above examples, the number of stars in the Adromeda Galaxy can be written as: $2.0 \times 100,000,000,000$

It is that large number, 100,000,000,000 which causes the problem. But that is just a multiple of ten. In fact it is ten times itself eleven times:

$$10 \times 10 \times 10 \times 10 \times 10 \times 10 \times 10 \times 10 \times 10 \times 10 \times 10 = 100,000,000,000$$

A more convenient way of writing 100,000,000,000 is 10^{11} .

The small number to the right of the ten is called the "exponent," or the "power of ten." *It represents the number of zeros that follow the 1.*

Though we think of zero as having no value, zeroes can make a number much bigger or smaller. Think about the difference between R10 and R100. Any one who has balanced a checkbook knows that one zero can make a big difference in the value of the number. So we would write 200,000,000,000 in scientific notation as: 2.0×10^{11}

This number is read as follows: "two point zero times ten to the eleventh."

This means that we have moved the decimal point eleven places to the left, until we only have one digit (not a 0) before it (the decimal point).

As we said above, the exponent refers to the number of zeros that follow the 1. So:

- ♦ $10^1 = 10$;
- ♦ $10^2 = 100$;
- ♦ $10^3 = 1,000$, and so on.

Similarly, $10^0 = 1$, since the zero exponent means that no zeros follow the 1.

Negative exponents indicate negative powers of 10, which are expressed as fractions with 1 in the numerator (on top) and the power of 10 in the denominator (on the bottom). So:

- ♦ $10^{-1} = 1/10$;
- ♦ $10^{-2} = 1/100$;
- ♦ $10^{-3} = 1/1,000$, and so on.

This allows us to express other small numbers this way. For example:

EXAMPLE:

- ♦ $2.5 \times 10^{-3} =$
- ♦ $2.5 \times 1/1,000 =$
- 0.0025**

You will see that we now have 10^{-3} . This means that we have been moving the decimal point to the right, until we only have one digit (not a 0) before it (the decimal point).

EXAMPLE:

1

$$99,1$$

$$= 9,91 \times 10^1$$

EXAMPLE:

2

$$3\,516\,000\,000$$

$$= 3,516 \times 10^9$$

EXAMPLE:

3

$$0,001$$

$$= 1 \times 10^{-3}$$

EXAMPLE:

4

$$0,000\,045\,9$$

$$= 4,59 \times 10^{-5}$$

When we convert from scientific to decimal notation, we move the decimal point in the opposite direction, i.e. if the exponent is *positive*, we move the decimal point to the *right*, but if the exponent is *negative*, we move the decimal point to the *left*.

EXAMPLE:

1

$$9.65 \times 10^3$$

$$= 9\,650$$

EXAMPLE:

2

$$7.2 \times 10^{15} \\ = 7\,200\,000\,000\,000\,000$$

EXAMPLE:

3

$$7.15 \times 10^{-2} \\ = 0.0715$$

EXAMPLE:

4

$$1.32 \times 10^{-6} \\ = 0.000\,001\,32$$

8. Decimal System

You must have a good understanding of the metric system (SI) in order to be able to convert from one unit to the other.

Metric units are based on *length* in **metres**, *mass* in **grams** and *volume* in **litres**. The following prefixes are commonly used:

- ♦ Milli - a thousandth ($\frac{1}{1000}$)
- ♦ Centi - a hundredth ($\frac{1}{100}$)
- ♦ Kilo - one thousand ($\frac{1000}{1}$)

Here is a summary of conversions between different units in the metric system:

		Big units to small units	Small units to big units
		→	←
Km	↔ m	$\times 1\,000$	$\div 1\,000$
m	↔ cm	$\times 100$	$\div 100$
m	↔ mm	$\times 1\,000$	$\div 1\,000$
cm	↔ mm	$\times 10$	$\div 10$
kg	↔ g	$\times 1\,000$	$\div 1\,000$
g	↔ mg	$\times 1\,000$	$\div 1\,000$
kl	↔ l	$\times 1\,000$	$\div \div 1\,000$
l	↔ ml	$\times 1\,000$	$\div 1\,000$

EXAMPLE:

1

Convert 2,35 m to mm. According to the table, multiply by 1 000.

Solution:

$$2,35 \text{ m} \times 1\,000 = 2\,350 \text{ mm}$$

EXAMPLE:

2

Convert 356 g to kg. According to the table divide by 1 000.

Solution:

$$356 \text{ g} \div 1\,000 = 0,356 \text{ g}$$

Here are some units that we very seldom see or even hear of:

- ♦ 1 metre = 10 *decimetres*
 = 100 centimetres
 = 1 000 millimetres
 = 1 000 000 *micrometres* (10^6)
 = 1 000 000 000 *nanometres* (10^9)
 = 1 000 000 000 000 *picometres* (10^{12})

For measures longer than a metre:

- ♦ 10 metres = 1 *decametre*
 100 metres = 1 *hectometre*
 1 000 = 1 kilometre
 1 000 000 = 1 *megametre* ($1\text{m} = 10^{-6}$ *megametre*)
 1 000 000 000 = 1 *gigametre* ($1\text{m} = 10^{-9}$ *gigametre*)

9. Converting from SI to Imperial System and vice versa

9.1 Conversion tables:

Length/Distance:

Imperial unit	Metric (SI) unit	Metric (SI) unit	Imperial unit
Inch	2.54 centimeters	Centimeter	0.39 inches
Foot	30.48 centimeters	Meter	3.28 feet
Yard	0.941 meters	Meter	1.09 yard
Mile	1.61 kilometers	Kilometer	0.62 miles

EXAMPLE:

1

4 miles
 = 4×1.61
 = 6.44 km

EXAMPLE:

2

6 cm
 = 6×0.39
 = 2.34 inches

UNIT STANDARD 9010: Demonstrate an understanding of the use of different number bases and measurement units and an awareness of error in the context of relevant calculations.

Mass/Weight:

Imperial unit	Metric (SI) unit	Metric (SI) unit	Imperial unit
Ounce (weight)	28.35 grams	Gram	0.035 ounces
Pound	0.45 kilograms	Kilogram	2.21 pounds
UK ton (2240 pounds)	1.02 metric tons	Metric ton (1000kg.)	0.98 UK tons
US tons (2000 pounds)	0.91 metric tons	Metric ton (1000kg.)	1.10 US tons



1

$$\begin{aligned} &2 \text{ pounds} \\ &= 2 \times 0.45 \\ &= 0.9 \text{ kg} \end{aligned}$$



2

$$\begin{aligned} &75 \text{ grams} \\ &= 75 \times 0.035 \\ &= 2.625 \text{ ounces} \end{aligned}$$

Temperature:

$$^{\circ}\text{C} = (^{\circ}\text{F} - 32) \div 1.8$$



1

$$\begin{aligned} &(68^{\circ}\text{F} - 32) \div 1.8 \\ &= (36) \div 1.8 \\ &= 20^{\circ}\text{C} \end{aligned}$$

$$^{\circ}\text{F} = (^{\circ}\text{C} \times 1.8) + 32$$



1

$$\begin{aligned} &(20^{\circ}\text{C} \times 1.8) + 32 \\ &= (36) + 32 \\ &= 68^{\circ}\text{F} \end{aligned}$$

Volume:

Imperial unit	Metric (SI) unit	Metric (SI) unit	Imperial unit
Teaspoon (UK)	5.92 millimeters	Milliliter	0.17 teaspoons (UK)
Teaspoon (US)	4.93 millimeters		0.20 teaspoons (US)
Tablespoon (UK)	17.76 millimeters	10 Milliliter	0.56 teaspoons (UK)
Tablespoon (US)	14.79 millimeters		0.68 teaspoons (US)
Fluid ounce (UK)	28.41 millimeters	100 Milliliter	3.52 fluid ounces (UK)
Fluid ounce (US)	29.57 millimeters		3.38 fluid ounces (US)
Pint (UK)	0.57 liters	Liter	1.76 pints (UK)
Pint (US)	0.47 liters		2.11 pints (US)
Quart (UK)	1.14 liters		0.88 quarts (UK)
Quart (US)	0.95 liters		1.06 quarts (US)
Gallon (UK)	4.55 liters		0.22 gallon (UK)
Gallon (US)	3.79 liters		0.26 gallon (US)

EXAMPLE:

1

5 pints

$= 5 \times 0.47$

$= 2.35$ liters

EXAMPLE:

2

30 milliliters

$= 30 \times 0.68$

$= 20.4$ tablespoons

10. Conversion factors

Here are the conversion factors, which make conversions much easier. All you need is a calculator!

Units	Formulas
Length	
Inches to Millimeters	x 25.4
Millimeters to inches	x 0.0394
Feet to Meters	x 0.305
Meters to Feet	x 3.281
Yards to Meters	x 0.914
Meters to Yards	x 1.094
Miles to Kilometers	x 1.609
Kilometers to Miles	x 0.621
Nautical Miles to Miles	x 1.151
Miles to Nautical Miles	x 0.869
Volume	
UK Gallons to Liters	x 4.546
Liters to UK Gallons	x 0.220
UK Pints to Liters	x 0.568
Liters to UK Pints	x 1.760
Weight	
Ounces to Grams	x 28.349
Grams to Ounces	x 0.035
Kilograms to Pounds	x 2.205
Pounds to Kilograms	x 0.454
UK tons to Tonnes	x 1.016
Tonnes to UK Tons	x 0.984
Temperature	
Fahrenheit to Celsius	$(F - 32) \div 1.8$
Celsius to Fahrenheit	$(C \times 1.8) + 32$
Speed	
mph to km/h	x 1.609
km/h to mph	x 0.621

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[illegible]

UNIT STANDARD 9010: Demonstrate an understanding of the use of different number bases and measurement units and an awareness of error in the context of relevant calculations.



WORDS/ACRONYMS	EXPLANATION OF ACRONYMS
ASSESSMENT	"A structured process of gathering evidence and making judgments about an individual's performance in relation to registered national standards and qualifications"
ASSESSMENT CRITERIA	It describe the evidence that is needed to show that the learner has demonstrated the outcome correctly. A list of evidence to be supplied by the candidate as proof of his / her ability to integrate all of the competencies listed earlier in the standard.
COMPETENCE	Competence is the ability to do certain things in a range of different environments with knowledge and understanding about what is being done and the effect this has on others. It includes <u>Knowledge, skills, attitudes, values and ethics.</u>
CREDIT	Each unit standard is worth a number of credits which gives an indication of the size and scope of the unit standard, the standard is worth. (1 credit = 10 notional hours of learning)
CRITICAL CROSSFIELD OUTCOMES	critical cross field outcomes is to direct educational activities towards development of the learners within a social and economic environment.
EMBEDDED KNOWLEDGE	Body of theory essential for understanding and applying the standard. Principles, methods and techniques that would underpin good practice of this unit standard.
EVIDENCE	Evidence is a set of information or observations that provides proof of the learner's competence.
FORMATIVE ASSESSMENT	Aims to assist the learner to identify any knowledge/skills gaps when demonstrating his/her current competence. Diagnostic assessment supports learning and progress of the learner.
INTEGRATED ASSESSMENT	Integrated assessment refers to: • assessing a number of outcomes together • assessing a number of assessment criteria together • assessing a number of unit standards together • using a combination of assessment methods and instruments for an outcome/s • collecting naturally occurring evidence in the workplace • Acquiring evidence from other sources such as supervisor's reports, testimonials, portfolio of work previously done, logbooks, journals etc.
LEARNING ASSUMED TO BE IN PLACE	Building blocks (knowledge, skill and understanding) assumed to be in place
LEARNING PROGRAM	A "learning program is a set of learning activities in which the learner will be involved in working towards the achievement of one or more outcomes.
LEVEL	Each unit standard is registered on the NQF at a particular level based on the complexity of the outcomes and stringency of the standards.
NQF	National Qualifications Framework
RANGE STATEMENTS	The range statements indicate detailed requirements of the assessment criteria.

RE-ASSESSMENT	• The same method and instrument may be used, but, the task and materials should be different • The task and materials should, however, be of the same complexity and level as the previous ones. • In case the methods and instruments are changed it must be ensured that they are appropriate for the outcomes specified.
RPL ASSESSMENT	RPL is a way of recognizing what individuals already know and can do. The basic premise underlying it is that people, especially mature adults, learn many things inside and outside the formal structures of education and training and that irrespective of where, how and when learning was acquired, subject to assessment, it can be worthy of recognition and credit.
RPL	Recognition of prior learning
SASSETA	Safety & security sector education and training authority.
SAQA	SOUTH AFRICAN QUALIFICATION AUTHORITY
SPECIFIC OUTCOMES	An outcome is a statement that describes the required competency that must be demonstrated by the learner on successful completion of a training intervention. Key performance areas in the standard are given, the focus is <i>not</i> only on current competence, but also on future competence in this standard.
SUMMATIVE ASSESSMENT	Assessment used to record the level of competence achieved by a learner over a period of time, e.g. at the end of a program, term or year.
QUALIFICATIONS	Represent a planned combination of learning outcomes which has a defined purpose and intended to provide learners with applied competence and a basis for further learning
UNIT STANDARDS	the “building blocks” of qualifications. All qualifications are plotted on the National Qualifications Framework (NQF). “A statement of the outcomes that the learner must achieve to earn credit. Standards are nationally agreed and internationally comparable.”

UNIT STANDARD 9010: Demonstrate an understanding of the use of different number bases and measurement units and an awareness of error in the context of relevant calculations.



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Demonstrate an understanding of the use of different number bases and measurement units and an awareness of error in the context of relevant calculations

SAQA US ID	UNIT STANDARD TITLE		
9010	Demonstrate an understanding of the use of different number bases and measurement units and an awareness of error in the context of relevant calculations		
ORIGINATOR		REGISTERING PROVIDER	
SGB Math Literacy, Math, Math Sciences L 2 -4			
FIELD		SUBFIELD	
Field 10 - Physical, Mathematical, Computer and Life Sciences		Mathematical Sciences	
ABET BAND	UNIT STANDARD TYPE	NQF LEVEL	CREDITS
Undefined	Regular-Fundamental	Level 3	2
REGISTRATION STATUS	REGISTRATION START DATE	REGISTRATION END DATE	SAQA DECISION NUMBER
Reregistered	2008-01-22	2011-01-22	SAQA 0160/05
LAST DATE FOR ENROLMENT		LAST DATE FOR ACHIEVEMENT	
2012-01-22		2015-01-22	

This unit standard does not replace any other unit standard and is not replaced by any other unit standard.

PURPOSE OF THE UNIT STANDARD

This unit standard is designed to provide credits towards the mathematical literacy requirements of the NQF at level 2. The essential purposes of the mathematical literacy requirements are that, as the learner progresses with confidence through the levels, the learner will grow in:

- . An insightful use of mathematics in the management of the needs of everyday living to become a self-managing person
- . An understanding of mathematical applications that provides insight into the learner's present and future occupational experiences and so develop into a contributing worker
- . The ability to voice a critical sensitivity to the role of mathematics in a democratic society and so become a participating citizen.

People credited with this unit standard are able to:

Convert numbers between the decimal number system and binary number system

Work with numbers in different ways to express size/magnitude.

Demonstrate the effect of error in calculations.

LEARNING ASSUMED TO BE IN PLACE AND RECOGNITION OF PRIOR LEARNING

The credit value is based on the assumption that people starting to learn towards this unit standard are competent in Mathematical Literacy and Communications at NQF level 2.

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UNIT STANDARD RANGE

This unit standard covers:

Approximation in relation to the use of computing technologies, the distinction between exact and approximate answers in a variety of problem settings. More detailed range statements are provided for specific outcomes and assessment criteria as needed.

Specific Outcomes and Assessment Criteria:**SPECIFIC OUTCOME 1**

Convert numbers between the decimal number system and the binary number system.

OUTCOME RANGE

This outcome includes the need to:

Perform addition and subtraction of positive whole numbers in binary up to 100002 (16 in decimal).

Demonstrate understanding of the mathematical relationships and principles involved in the computations.

ASSESSMENT CRITERIA**ASSESSMENT CRITERION 1**

1. Conversion between binary and decimal numbers is done correctly.

ASSESSMENT CRITERION 2

2. Basic addition and subtraction calculations in the binary number system are done correctly.

ASSESSMENT CRITERION RANGE

Using positive whole numbers up to the 16 in decimal.

ASSESSMENT CRITERION 3

3. Practical applications of the decimal and binary system are explained correctly.

SPECIFIC OUTCOME 2

Work with numbers in different ways to express size and magnitude.

OUTCOME RANGE

This outcome includes the need to:

Use scientific notation for small and large numbers.

ASSESSMENT CRITERIA**ASSESSMENT CRITERION 1**

1. The prefixes indicating magnitude in measurements are correctly related to the decimal system.

ASSESSMENT CRITERION RANGE

From Giga to Pico (10^{12} to 10^{-12})

ASSESSMENT CRITERION 2

2. Conversions between related units in different measurement systems are correctly applied in real-life contexts.

ASSESSMENT CRITERION RANGE

SI to Imperial; Degrees F to degrees C.

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SPECIFIC OUTCOME 3

Demonstrate the effect of error in calculations.

OUTCOME RANGE

This outcome includes the need to:

Work with rational and irrational numbers.

Explore repeating decimals and convert them to common fraction form

Use scientific notation for small and large numbers.

ASSESSMENT CRITERIA**ASSESSMENT CRITERION 1**

1. Symbols for irrational numbers such as $\sqrt{7}$ and $\sqrt{42}$ are left in formulae or steps to calculations except where approximations are required.

ASSESSMENT CRITERION 2

2. Descriptions are provided of the effect of rounding prematurely in calculations.

ASSESSMENT CRITERION 3

3. The desired degree of accuracy is determined in relation to the practical context.

ASSESSMENT CRITERION 4

4. The final value of a calculation is expressed in terms of the required unit.

UNIT STANDARD ACCREDITATION AND MODERATION OPTIONS

Providers of learning towards this unit standard will need to meet the accreditation requirements of the GENFETQA.

Moderation Option:

The moderation requirements of the GENFETQA must be met in order to award credit to learners for this unit standard.

UNIT STANDARD ESSENTIAL EMBEDDED KNOWLEDGE

The following essential embedded knowledge will be assessed through assessment of the specific outcomes in terms of the stipulated assessment criteria. Candidates are unlikely to achieve all the specific outcomes, to the standards described in the assessment criteria, without knowledge of the listed embedded knowledge. This means that the possession or lack of the knowledge can be inferred directly from the quality of the candidate's performance against the standards.

Number systems and rational and irrational numbers
Estimation and approximation
Scientific notation

Critical Cross-field Outcomes (CCFO):**UNIT STANDARD CCFO COLLECTING**

Collect, analyse, organise and critically evaluate information:
Gather, organise, and interpret numerical information.

UNIT STANDARD CCFO COMMUNICATING

Communicate effectively:
Use everyday language and mathematical language to describe relationships, processes and problem solving methods.

UNIT STANDARD 9010: Demonstrate an understanding of the use of different number bases and measurement units and an awareness of error in the context of relevant calculations.

